The inventory gerbe couples on the torus

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Abstract

A supply chain, abstracted to a small category $\mathcal{C}$, carries fungible inventory as local stock-data glued along logistics operations up to a relabelling group $K = \mathrm{Aut}F$ of the stock fibre F; the gluing datum is an $\mathrm{Aut}F$-gerbe over the nerve $N(\mathcal{C})$, and following Giraud its obstruction to a strict global inventory is *staged*: a nonabelian band $\beta \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ must vanish before an abelian class $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(K))$ is even defined. We ask whether the stages are independent. Over a one-dimensional base they are, but over a two-dimensional one they need not be. We prove a general *δ-commutator lemma*: for an abelian-kernel extension of coefficient groups and a band $\rho\colon \mathbb{Z}^2 \to Q$ on the torus, the connecting map is given exactly by the commutator of any two lifts, $\delta(\rho) = [[A, B]]$, in the coinvariants $H^2(T^2; M)$. Consequently a structure group couples on T^2 – a nonzero band forcing a nonzero degree-two class – if and only if some commuting outer pair admits no commuting lift, which forces $\mathrm{Aut}K \to \mathrm{Out}K$ to be non-split. We then locate the phenomenon precisely: for the n-lot ring $K = D_n$ the inventory class is essentially nonzero exactly when $4 \mid n$, and across an enumerated family of order-16 nilpotency-class-two groups only $D_4 \times C_2$ and $Q_8 \times C_2$ couple. Coupling is a rare, identifiable feature of gauge-structured stock, not of mere interchangeability.

1 Introduction

The problem. A supply chain is a small category $\mathcal{C}$: objects are nodes (warehouses, plants, ledgers), morphisms are logistics operations, composition is sequencing. Inventory is a datum glued along the morphisms. When two local inventory charts are comparable only *up to a relabelling* of stock states – that is, up to the automorphism group $K := \mathrm{Aut}F$ of the stock fibre F – the gluing datum is no longer a presheaf but an $\mathrm{Aut}F$-gerbe over the nerve $N(\mathcal{C})$. A *strict global inventory* is a global section: a single coherent choice of trivialization valid across the whole chain. The question with economic teeth is: *when does one exist?* A failure is a compositional defect – stock-keeping that cannot be reconciled globally no matter how the local books are kept.

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The staged obstruction. In two companion notes we showed that the obstruction is not a single cohomology class but a *tower*, exactly matching Giraud's three-stratum classification of nonabelian gerbes [4, 1]. The relabelling group $K = \mathrm{Aut}F$ sits in two abelian-kernel extensions that share their middle term $\mathrm{Inn}K$:

$$\text{(II)}\quad 1 \to Z(K) \to K \to \mathrm{Inn}K \to 1, \qquad \text{(I)}\quad 1 \to \mathrm{Inn}K \to \mathrm{Aut}K \xrightarrow{q} \mathrm{Out}K \to 1. \tag{1}$$

The *band* is the outer transport $\beta = [q\circ\alpha] \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ read off extension (I); it is a genuinely nonabelian datum, and it must vanish before the abelian *inventory class* $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(K))$ – the stratum-three class of extension (II) – is even well-defined. A strict global inventory exists iff *both* vanish (Corollary 5, the *floor*).

The gap. The floor leaves a question the companion notes flagged but did not answer: are the two stages *independent*, or can one force the other? On a base of cohomological dimension one – e.g. a single reversible operation, $\mathcal{C} = B\mathbb{Z}$, $N(\mathcal{C}) \simeq S^1$ – they are forced apart, because $H^2(S^1; -) = 0$ annihilates every degree-two class; every band is strictifiable and the stages stack freely. Among the *free* bases $B(\mathbb{Z}^n)$ the first that can hold a degree-two class is the torus $T^2 \simeq N(B\mathbb{Z}^2)$: two commuting reversible operations. (Over a *torsion* base, coupling can already occur at rank one; see §7.) There the question is open on its face.

The result. We close it with a single computation, valid for every abelian-kernel extension of coefficient groups. Write the band as a homomorphism $\rho\colon \mathbb{Z}^2 \to Q$ into the relevant quotient ($Q = \mathrm{Out}K$ or $Q = \mathrm{Inn}K$), and let M be the (abelian) kernel. Because $\mathbb{Z}^2$ is abelian, a band is exactly a *commuting pair* of classes $\rho(s), \rho(t) \in Q$.

Theorem 1 (δ-commutator lemma; see Theorem 8)**.** *Let $1 \to M \to E \xrightarrow{\pi} Q \to 1$ be an extension of groups with M abelian, and let $\rho\colon \mathbb{Z}^2 \to Q$ be a band with lifts $A, B \in E$ of $\rho(s), \rho(t)$. Then $w := [A, B] = ABA^{-1}B^{-1}$ lies in M, and under the canonical identification $H^2(T^2; M) = M_{\mathbb{Z}^2}$ with the coinvariants, the connecting map is*

$$\delta(\rho) = [\,w\,] \in M_{\mathbb{Z}^2},$$

independently of the lifts. In particular $\delta(\rho) = 0$ iff the band admits commuting lifts, iff ρ lifts to a homomorphism $\mathbb{Z}^2 \to E$.

The method is deliberately elementary – a Schreier-style computation in the discrete, nonabelian register, requiring no gerbe machinery beyond the classical dictionary between H^2 and extensions [2, 3]. The one non-classical step is an explicit comparison chain map between the Koszul and bar resolutions of $\mathbb{Z}$ over $\mathbb{Z}[\mathbb{Z}^2]$, sending the fundamental 2-cell to the antisymmetric expression $\varphi_2(e_{st}) = [s\,|\,t] - [t\,|\,s]$; this turns the slogan "evaluate the obstruction cocycle on the torus relator $sts^{-1}t^{-1}$" into the identity $\delta(\rho) = [f(s,t) - f(t,s)]$ and pins the class to the commutator.

Consequences. Theorem 1 gives an immediately checkable *coupling criterion* (Corollary 10): a structure group K couples on T^2 – a nonzero band forcing a nonzero degree-two class – iff some commuting pair in the quotient admits no commuting lift, and this forces the governing extension to be *non-split*. Two sharp locations follow.

- *Sharpness* (Theorem 12). For the n-lot ring $K = \mathrm{Aut}(C_n) = D_n$ the inventory class $[\mathfrak{g}]$ is essentially nonzero iff the central extension (II) is non-split iff $4 \mid n$. The six-lot ring $D_6 \cong \mathbb{Z}/2 \times S_3$ – nonabelian with nontrivial centre, yet split – is the witness that "nonabelian with nontrivial centre" is necessary but not sufficient.

- *Rarity* (§6). Across an enumerated battery of order-16 nilpotency-class-two groups, only $D_4 \times C_2$ and $Q_8 \times C_2$ couple. Non-splitting alone is genuinely weaker than coupling: M_{16} and $C_4 \rtimes C_4$ need not split yet carry no coupled band.

Compositional failure of global stock-keeping is therefore a feature of a thin, identifiable slice of structure groups – gauge-structured stock with non-split automorphism tower – and is invisible to fungibility.

Status and provenance. This is a consolidation at honest grades, assembled from four proof files; the only new theorem is the general δ-commutator lemma. We record the grade of each ingredient ("lean-verified" means a finite core is machine-checked in Lean 4/Mathlib).

Result	Location	Grade	Lean
Floor: strict $\iff \beta = 0 \wedge [\mathfrak{g}] = 0$	Cor. 5	proved	—
δ-commutator lemma; coupling criterion	Thm. 8, Cor. 10	proved	—
Coupling witness $D_4 \times C_2$ on T^2	§4	computed	finite core
Sharp line: $[\mathfrak{g}] \neq 0 \iff 4 \mid n$	Thm. 12	proved	finite core
Rarity over the order-16 battery	§6	computed	—

Outline. Section 2 fixes the inventory gerbe, its two extensions, and the floor. Section 3 proves the δ-commutator lemma and the coupling criterion. Section 4 gives the $D_4 \times C_2$ witness on T^2. Section 5 proves the sharp $4 \mid n$ line. Section 6 reports the rarity enumeration. Section 7 draws the consequences for compositional correctness.

2 The inventory gerbe and its staging

The gerbe. Let $\mathcal{C}$ be a small category and $N(\mathcal{C})$ its nerve. An *inventory fibre* F is the local type of a stock-state at a node; its automorphism group $K := \mathrm{Aut}F$ is the *relabelling group* of fungible lots. When local inventory charts are comparable only up to K, the gluing data form an $\mathrm{Aut}F$-gerbe over $N(\mathcal{C})$: a transport $\alpha\colon \mathrm{Mor}\mathcal{C} \to \mathrm{Aut}K$ (re-trivializing a chart conjugates the next transport), functorial *up to inner automorphism* of K, together with higher coherence valued in $Z(K)$. We use throughout the inner/outer and centre sequences of K, which are precisely the two extensions (1); note that $\mathrm{Inn}K$ is at once the quotient $K/Z(K)$ of (II) and the kernel of (I), so the two extensions genuinely stack.

Cohomology over $N(\mathcal{C})$ is simplicial cohomology of the nerve; for a group Q of constant coefficients, a class in nonabelian $H^1(N(\mathcal{C}); Q)$ is a functor $\mathcal{C} \to BQ$ modulo object-relabelling, pointed by the trivial functor [4]. When $\mathcal{C} = B\Gamma$ for a group Γ, $N(\mathcal{C}) = B\Gamma$ computes ordinary group cohomology of Γ; the two bases we need are $\mathcal{C} = B\mathbb{Z}$ ($N \simeq S^1$) and $\mathcal{C} = B\mathbb{Z}^2$ ($N \simeq T^2$).

The band. Projecting the transport by $q\colon \mathrm{Aut}K \to \mathrm{Out}K$ kills the up-to-inner slack and yields a strict functor $\rho := q \circ \alpha\colon \mathcal{C} \to B(\mathrm{Out}K)$.

Definition 2. The *band* of the inventory gerbe is $\rho = q \circ \alpha$, with class $\beta := [\rho] \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ (the Giraud stratum-two invariant). The band is *trivial* if $\beta = 0$.

That ρ is a genuine cocycle regardless of the inner slack is immediate: if $\alpha_{gf} = \iota\, \alpha_g \alpha_f$ with $\iota \in \mathrm{Inn}K = \ker q$, then $\rho(gf) = q(\alpha_{gf}) = \rho(g)\rho(f)$ because $q(\iota) = e$.

The inventory class. Once the band is trivial, the companion note constructs the stratum-three datum. Choose section-lifts $T_f \in K$ of the inner transport $\bar{\rho}\colon \mathcal{C} \to B(\mathrm{Inn}K)$; the defect $T_g T_f = \mathfrak{g}(g,f) \cdot T_{gf}$ lands in $\ker(K \to \mathrm{Inn}K) = Z(K)$ because both sides agree in $\mathrm{Inn}K$. Associativity in K, with $Z(K)$ central, makes $\mathfrak{g}$ a normalized 2-cocycle, and changing the section changes it by a coboundary.

Definition 3. The *inventory class* is $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(K))$, the pullback along the inner transport of the class of the central extension (II). Its coefficient group is $Z(K)$, with trivial action (inner automorphisms fix the centre pointwise).

The floor. The two strata assemble into a single necessary-and-sufficient criterion, proved in the companion band-boundary note. First we fix *what* is to be globalized, retaining the gerbe's 2-cell datum rather than only its outer shadow.

Definition 4 (global inventory, D2)**.** A *global inventory* is a pair (μ, T) in which $\mu\colon \mathrm{Ob}\mathcal{C} \to \mathrm{Aut}K$ is a gauge and $T\colon \mathcal{C} \to BK$ is a *strict* functor, *retaining* the gerbe's 2-cell datum $k_{g,f} \in K$ (so that $\alpha_{gf} = \mathrm{conj}(k_{g,f}) \cdot \alpha_g \alpha_f$), subject to

$$\mu(y) \cdot \alpha(f) \cdot \mu(x)^{-1} = \mathrm{conj}(Tf) \qquad (f\colon x \to y).$$

A *strict global inventory* is such a pair. It is the full stratum-three globalization, not merely the bare outer datum.

Corollary 5 (the floor)**.** *Under Definition 4, a strict global inventory exists if and only if both strata vanish:* (i) $\beta = 0 \in H^1(N(\mathcal{C}); \mathrm{Out}K)$*, and* (ii) $[\mathfrak{g}] = 0 \in H^2(N(\mathcal{C}); Z(K))$*, where* (ii) *is well-defined once* (i) *holds.*

Proof sketch. A strict global inventory (μ, T) in the sense of Definition 4 has its gauge μ projecting by q to a trivialization of ρ, so (i) is necessary. Given (i), one may choose α inner-valued (band-triviality is exactly "inner-valued up to an outer relabelling"), whereupon – using the retained 2-cell datum $k_{g,f}$ – the gerbe is the honest gerbe of lifts of $\bar{\rho}$ through (II), neutral iff $[\mathfrak{g}] = 0$ by the central-extension lifting theorem [2, IV.3]. Full details, including the fungibility corollary (for an n-element commodity fibre $\mathrm{Out}(S_n) = 1$ for every n except $n = 6$, and $Z(S_n) = 1$ for $n \geq 3$ while S_2 is abelian so $[\mathfrak{g}] = 0$ there as well [5, Thms 7.5–7.7]; hence both strata vanish for *all* n and the only fungible exception is the band-only $n = 6$, $\mathrm{Out}(S_6) = \mathbb{Z}/2$), are in the band-boundary note [commit `1c9affb`]. □

Remark 6 (D1 versus D2)**.** Under the naive definition (D1) that drops the retained 2-cell datum $k_{g,f}$, the equivalence *fails*: the secondary stage is then a class in $H^1(\mathcal{C}; \mathrm{Inn}G)$ modulo $H^0(\mathcal{C}; \mathrm{Out}G)$, not in H^2. Rick's counterexample is $\mathcal{C} = B\mathbb{Z}$ with $G = S_3$ and $\alpha = \mathrm{conj}((01))$: there $\beta = 0$ and $[\mathfrak{g}] = 0$, yet no gauge μ strictifies the transport (Rick, band-boundary review, 2026-10-12). Retaining $k_{g,f}$, as in Definition 4, is exactly what restores the clean H^2 floor.

The floor says the two obstructions *stack*: clear the band, then clear the class. The rest of the paper asks whether clearing them is independent – and finds that on T^2 it is not.

3 The δ-commutator lemma

Fix an extension of groups

$$1 \longrightarrow M \xrightarrow{\iota} E \xrightarrow{\pi} Q \longrightarrow 1 \tag{2}$$

with M *abelian*. Then M is a Q-module by conjugation (π-equivariantly, the action descending because M is abelian), and a band $\rho\colon \mathbb{Z}^2 \to Q$ makes M a $\mathbb{Z}^2$-module: writing $\sigma := \rho(s)\cdot(-)$, $\tau := \rho(t)\cdot(-) \in \mathrm{Aut}M$ (commuting, as $\mathbb{Z}^2$ is abelian). Because $T^2 = B\mathbb{Z}^2$ is a $K(\mathbb{Z}^2,1)$ and M is abelian, $H^\bullet(T^2;M) = H^\bullet(\mathbb{Z}^2;M)$ is ordinary group cohomology. We will apply this with (E,Q,M) equal to $(\mathrm{Aut}K, \mathrm{Out}K, \mathrm{Inn}K)$ – the band-strictification extension (I), which needs K of nilpotency class ≤ 2 so that $\mathrm{Inn}K$ is abelian – and to $(K, \mathrm{Inn}K, Z(K))$ – the central extension (II), where $M = Z(K)$ is abelian with trivial action for every K.

3.1 The target group and the comparison map

Let $R = \mathbb{Z}[\mathbb{Z}^2] = \mathbb{Z}[s^{\pm1}, t^{\pm1}]$. The Koszul resolution of the trivial module $\mathbb{Z}$,

$$0 \to R\langle e_{st}\rangle \xrightarrow{\partial_2^K} R\langle e_s, e_t\rangle \xrightarrow{\partial_1^K} R\langle 1\rangle \to \mathbb{Z} \to 0,$$

with $\partial_1^K e_s = s-1$, $\partial_1^K e_t = t-1$, and $\partial_2^K e_{st} = (s-1)e_t - (t-1)e_s$, computes, after $\mathrm{Hom}_R(-,M)$, the complex $M \xrightarrow{d^1} M^2 \xrightarrow{d^2} M$ with $d^2(m_1,m_2) = (\sigma-1)m_2 - (\tau-1)m_1$, whence

$$H^2(\mathbb{Z}^2;M) = \mathrm{coker} d^2 = M\big/\big((\sigma-1)M + (\tau-1)M\big) = M_{\mathbb{Z}^2}, \tag{3}$$

the coinvariants. To read a bar cocycle in these coordinates we need the comparison map to the bar resolution $B_\bullet$ (free on normalized tuples $[g_1|\cdots|g_n]$, with $\partial_2^B[g|h] = g[h] - [gh] + [g]$).

Lemma 7. *The R-linear maps $\varphi_0(1) = [\,]$, $\varphi_1(e_s) = [s]$, $\varphi_1(e_t) = [t]$, and*

$$\varphi_2(e_{st}) = [s\,|\,t] - [t\,|\,s]$$

form a chain map $K_\bullet \to B_\bullet$ lifting $\mathrm{id}_{\mathbb{Z}}$. Consequently, for any bar 2-cocycle $f\colon (\mathbb{Z}^2)^2 \to M$,

$$\varphi^*[f] = [\,f(s,t) - f(t,s)\,] \in M_{\mathbb{Z}^2}. \tag{4}$$

Proof. Only the top degree needs checking. On one side $\varphi_1\partial_2^K e_{st} = (s-1)[t] - (t-1)[s]$. On the other,

$$\partial_2^B\big([s|t] - [t|s]\big) = \big(s[t] - [st] + [s]\big) - \big(t[s] - [ts] + [t]\big) = (s-1)[t] - (t-1)[s],$$

the terms $-[st]+[ts]$ cancelling because $st = ts$ in $\mathbb{Z}^2$. As a chain map between free resolutions of $\mathbb{Z}$ lifting the identity, φ induces the canonical isomorphism $\varphi^*\colon H^2_{\mathrm{bar}}(\mathbb{Z}^2;M) \xrightarrow{\sim} M_{\mathbb{Z}^2}$, unique up to homotopy, and (4) is its value on $[f] = [f\circ\varphi_2]$. □

Equation (4) is the precise form of "evaluate the obstruction cocycle on the fundamental 2-cell": the cell e_{st} maps to the antisymmetric expression $[s|t] - [t|s]$, the algebraic shadow of the torus relator $sts^{-1}t^{-1}$.

3.2 The lemma

Theorem 8 (δ-commutator lemma)**.** *With the extension (2) and band $\rho\colon \mathbb{Z}^2 \to Q$ as above, let $A, B \in E$ be any lifts of $\rho(s), \rho(t)$. Then $w := [A,B] = ABA^{-1}B^{-1} \in \ker\pi = M$, and*

$$\delta(\rho) = [\,w\,] \in M_{\mathbb{Z}^2} = H^2(T^2;M).$$

The class $[w]$ is independent of the choice of lifts, and

$$\delta(\rho) = 0 \iff \exists \text{ commuting lifts } A, B \iff \rho \text{ lifts to a homomorphism } \mathbb{Z}^2 \to E.$$

Here δ is the connecting map of the pointed sequence $H^1(T^2;E) \xrightarrow{\pi_*} H^1(T^2;Q) \xrightarrow{\delta} H^2(T^2;M)$; concretely, $\delta(\rho)$ is the class of the pullback extension $E_\rho = E \times_Q \mathbb{Z}^2 = \{(a,\gamma) : \pi(a) = \rho(\gamma)\}$ of $\mathbb{Z}^2$ by M, which splits iff ρ lifts to a homomorphism [2, IV.3–4]. Thus the vanishing criterion is classical; the content is the *value* $[w]$.

Proof. *(i)* $\pi(w) = [\rho(s), \rho(t)] = 1$ because $\rho(s), \rho(t)$ are images of the commuting generators s, t under a homomorphism; so $w \in \ker \pi = M$.

(ii) Work in E_ρ and choose lifts $\hat{s} = (A, s)$, $\hat{t} = (B, t)$; then $[\hat{s}, \hat{t}] = ([A, B], 1) = (w, 1)$, the commutator of the two lifts inside $\ker p \cong M$, where $p(a, \gamma) = \gamma$. Define the set-section $u(s^i t^j) := \hat{s}^{\,i} \hat{t}^{\,j}$ (every element of $\mathbb{Z}^2$ is uniquely $s^i t^j$; $u(1) = 1$, $p \circ u = \mathrm{id}$). Its factor set $f(g,h) := u(g)u(h)u(gh)^{-1} \in M$ is a normalized 2-cocycle representing $[E_\rho]$ [2, IV.3]. Using $ts = st$, so $u(ts) = u(st) = \hat{s}\hat{t}$,

$$f(s,t) = \hat{s}\hat{t}(\hat{s}\hat{t})^{-1} = 1, \qquad f(t,s) = \hat{t}\hat{s}(\hat{s}\hat{t})^{-1} = \hat{t}\hat{s}\hat{t}^{-1}\hat{s}^{-1} = [\hat{s}, \hat{t}]^{-1} = w^{-1}.$$

Additively $f(s,t) = 0$, $f(t,s) = -w$, so by (4), $\delta(\rho) = \varphi^*[f] = [0 - (-w)] = [w] \in M_{\mathbb{Z}^2}$.

(iii) Lift-independence. Any other lifts are $A' = Au$, $B' = Bv$ with $u, v \in M$ (two lifts of the same class differ by $\ker \pi = M$). Moving every M-letter to the left in $[Au, Bv]$ – each picking up the action of the automorphisms to its left – gives, additively,

$$[Au, Bv] - [A, B] = \tau(\sigma - 1)v - \sigma(\tau - 1)u. \tag{5}$$

Since σ, τ commute with $\sigma - 1$ and $\tau - 1$, the right-hand side lies in $(\sigma - 1)M + (\tau - 1)M$; hence $[w]$ is well-defined in the coinvariants (3).

(iv) Vanishing. By the extension dictionary, $\delta(\rho) = 0$ iff E_ρ splits iff ρ lifts to a homomorphism; and by (5), commuting lifts exist iff some (u, v) makes $[Au, Bv] = 0$ iff $[w] = 0$ in $M_{\mathbb{Z}^2}$. □

Remark 9 (criterion versus slogan)**.** The vanishing criterion is "no commuting lifts", i.e. $w \notin (\sigma - 1)M + (\tau - 1)M$ – strictly weaker than $w \neq 0$ in M when the action is nontrivial. When $\sigma = \tau = \mathrm{id}$ (trivial action, as for every witness in §4–6) the coboundaries vanish, $M_{\mathbb{Z}^2} = M$, and the criterion is the clean $w = [A, B] \neq 0$. The identity (5) was additionally verified element-wise in E across a battery of class-two groups, including *nontrivial-action* pairs that no single trivial-action witness reaches.

3.3 The coupling criterion

Say K *couples on* T^2 *via* (2) if some band $\rho\colon \mathbb{Z}^2 \to Q$ has $\delta(\rho) \neq 0$ (then automatically $\beta = [\rho] \neq 0$, since the trivial band has $w = [\mathrm{id}, \mathrm{id}] = 0$). Theorem 8 converts this into a finite search.

Corollary 10 (coupling criterion)**.** *K couples on T^2 via (2) iff there exist commuting $\bar{A}, \bar{B} \in Q$ whose lifts to E never commute. In particular coupling forces (2) to be* non-split*; non-splitting is necessary but not sufficient.*

Proof. The equivalence is Theorem 8(iv). If (2) splits by $\theta\colon Q \to E$, then $\theta \circ \rho$ is a homomorphic lift of every band, so $\delta \equiv 0$; contrapositively coupling forces non-split. Necessity is not sufficiency: a band is a *commuting pair* of nontrivial classes, and a rank-one image never couples (since $H^2(\mathbb{Z}; -) = 0$); one needs the extension class to restrict nontrivially to some commuting $\mathbb{Z}^2 \to Q$, which can fail even when the class is nonzero – see §6. □

Because $S^1 = B\mathbb{Z}$ has $H^2(S^1; -) = 0$, the connecting map vanishes identically there: every band over a one-dimensional *free* base is strictifiable, and the two Giraud strata stack freely. Among the free bases $B(\mathbb{Z}^n)$ the torus is therefore the first where coupling can occur, and Corollary 10 says exactly when it does. This "first" is relative to the free bases only. Over a *torsion* base coupling already appears at rank one: $B(\mathbb{Z}/2)$ has nerve $\simeq \mathbb{RP}^\infty$ with $H^2(B(\mathbb{Z}/2); -) \neq 0$, and $K = D_4 \times C_2$ with the outer involution $a\colon (r \mapsto r,\ s \mapsto sc,\ c \mapsto r^2c)$, $a^2 = \mathrm{conj}(g) \neq 1$, admits no strict lift (Rick, band-boundary review, 2026-10-12). So the torus is the first *free* base, not the first base overall.

4 A coupled witness on the torus

We exhibit a structure group that couples, via both extensions (1) at once. The following is *computed*: a finite brute-force certificate over the full (machine-verified complete) automorphism group, with the arithmetic core independently re-checked in Lean.

The group. Take $K = D_4 \times C_2 = \langle r, s\rangle \times \langle c\rangle$ with $r^4 = s^2 = c^2 = 1$, $srs^{-1} = r^{-1}$, c central. Then K has nilpotency class two,

$$Z(K) = \{1, r^2, c, r^2c\} \cong (\mathbb{Z}/2)^2, \quad \mathrm{Inn}K = K/Z(K) \cong (\mathbb{Z}/2)^2 = \langle [r], [s]\rangle, \quad |\mathrm{Out}K| = 16,$$

and $|\mathrm{Aut}K| = 64$ (verified). The second central involution c – absent from D_4 and Q_8 alone – is what lets automorphisms mix the three central involutions r^2, c, r^2c and so breaks the splitting of (I).

The band (strictification coupling, extension (I)). Take

$$A\colon\ r \mapsto r^3,\ s \mapsto s,\ c \mapsto r^2c, \qquad B\colon\ r \mapsto r^3c,\ s \mapsto s,\ c \mapsto c.$$

Both are automorphisms, both are *outer*, both have order two in $\mathrm{Out}K$ and commute, so $\rho\colon \mathbb{Z}^2 \twoheadrightarrow \langle [A], [B]\rangle \cong (\mathbb{Z}/2)^2 \hookrightarrow \mathrm{Out}K$, $s \mapsto [A]$, $t \mapsto [B]$, is a nontrivial band: $\beta \neq 0$. Both act trivially on $\mathrm{Inn}K = K/Z(K)$ (e.g. $A(r) = r^3 \equiv r$, $A(c) = r^2c \equiv c \bmod Z$), so the action is trivial and $H^2(T^2; \mathrm{Inn}K) = \mathrm{Inn}K = (\mathbb{Z}/2)^2$. The commutator of lifts is

$$[A, B] = c_s \text{ (conjugation by } s), \qquad \delta(\rho) = [A, B] = [s] \neq 0 \in H^2(T^2; \mathrm{Inn}K) = (\mathbb{Z}/2)^2.$$

Exhaustively over all 16×16 inner adjustments (Au, Bv), none commute, so by Corollary 10 the band is *not* strictifiable: the stratum-two band forces the degree-two strictification class. The strata do not split.

The central coupling (extension (II)). The same group couples the inventory class. The inner band $s \mapsto [r]$, $t \mapsto [s]$ lifts to K with commutator $[r, s] = r^2 \in Z(K)$, so $[\mathfrak{g}] = [r^2] \neq 0 \in H^2(T^2; Z(K)) = (\mathbb{Z}/2)^2$ (trivial action). Thus $D_4 \times C_2$ realizes the coupling at both stages of the tower (1): $\mathrm{Out}K \leftarrow \mathrm{Aut}K$ (coeff $\mathrm{Inn}K$) and $\mathrm{Inn}K \leftarrow K$ (coeff $Z(K)$).

Remark 11 (grade and scope)**.** Both couplings are finite brute-force certificates (*computed*); the arithmetic core – that A, B are commuting outer automorphisms acting trivially on $\mathrm{Inn}K$, that $[A, B] = c_s \neq 0$, and that no inner adjustment makes the lifts commute – is machine-checked in Lean (*finite core lean-verified*). We do *not* claim $D_4 \times C_2$ is minimal over all groups of order ≤ 16; only that it is the smallest coupler in the enumerated battery of §6.

5 The sharp line: the n-lot ring

The witness shows coupling *can* occur. The n-lot ring pins down *exactly when* the inventory class (extension (II)) is essentially nonzero, as an arithmetic condition on n. Here the result is *proved*, with the finite split-test core lean-verified.

The structure group. Let $F(n)$ be n fungible lots arranged in a ring – the vertices of the n-cycle C_n with cyclic adjacency, $n \geq 3$. Its relabelling group is the dihedral group

$$K(n) = \mathrm{Aut}(C_n) = D_n = \langle r, s \mid r^n = s^2 = 1,\ srs^{-1} = r^{-1} \rangle \quad (\text{order } 2n).$$

Call the inventory class $[\mathfrak{g}]$ *essentially nonzero* if some base $(\mathcal{C}, \bar{\rho})$ with inner transport gives $[\mathfrak{g}] \neq 0$ – i.e. strict global inventory is obstructed for some supply chain on this fibre.

Theorem 12 (sharp n-lot criterion)**.** *For $K = D_n$,*

$$[\mathfrak{g}] \text{ essentially nonzero} \iff \text{the central extension (II) is non-split} \iff 4 \mid n.$$

Equivalently, strict global inventory on the n-lot ring is always rectifiable unless $4 \mid n$.

Proof. Centre and inner group. A rotation r^k is central iff $r^{2k} = 1$, giving $k = 0$ (all n) and $k = n/2$ (n even); a reflection is never central for $n \geq 3$. Hence $Z(D_n) = 1$ for n odd and $\langle r^{n/2} \rangle \cong \mathbb{Z}/2$ for n even, and $\mathrm{Inn}D_n = D_n/Z(D_n)$ is D_n (n odd) or $D_{n/2}$ (n even). For n odd, $Z = 1$, so (II) is $1 \to 1 \to D_n \to D_n \to 1$ and the coefficient group $Z(D_n) = 1$ makes $[\mathfrak{g}]$ *vacuously* zero; the question lives in the even case.

The split jump. Fix $n = 2m$, $m \geq 2$. A splitting of $1 \to \langle r^m \rangle \to D_{2m} \to D_m \to 1$ is a complement H to the central $Z = \langle r^m \rangle$; being of index two it is normal, so a complement makes $D_{2m} = Z \times H$ an internal direct product with $H \cong D_m$. Thus

$$\text{(II) splits} \iff D_{2m} \cong \mathbb{Z}/2 \times D_m.$$

Compare centres, an isomorphism invariant: $|Z(D_{2m})| = 2$ always, while $|Z(\mathbb{Z}/2 \times D_m)| = 2\,|Z(D_m)|$ equals 2 iff $|Z(D_m)| = 1$ iff m is odd. So for m even the two groups are non-isomorphic and (II) is non-split; for m odd an explicit complement exists, namely $H = \langle r^2, s \rangle \cong D_m$ (since m odd makes r^2 of order m and $r^m \notin \langle r^2 \rangle$, so $H \cap Z = 1$ and $|HZ| = 4m = |D_{2m}|$). Hence (II) is non-split iff n even and $n/2$ even, i.e. iff $4 \mid n$.

From splitting to the class. With trivial coefficients, $[\star] = 0$ in $H^2(\mathrm{Inn}K; Z(K))$ iff (II) splits ([2, IV.3]; [3]). Taking $\mathcal{C} = B(\mathrm{Inn}K)$, $\bar{\rho} = \mathrm{id}$ realizes $[\mathfrak{g}] = [\star]$, while for any $(\mathcal{C}, \bar{\rho})$, $[\mathfrak{g}] = \bar{\rho}^*[\star] = 0$ whenever $[\star] = 0$. So $[\mathfrak{g}]$ is essentially nonzero iff $[\star] \neq 0$ iff (II) non-split iff $4 \mid n$. □

Remark 13 (the D_6 reconciliation)**.** The six-lot ring is the witness that "nonabelian with nontrivial centre" is *necessary but not sufficient* for $[\mathfrak{g}] \neq 0$. D_6 is nonabelian with $Z(D_6) = \langle r^3 \rangle \cong \mathbb{Z}/2 \neq 1$, so it satisfies the naive hypothesis; but $6 = 2 \cdot 3$ with $m = 3$ odd, so by the split jump $D_6 \cong \mathbb{Z}/2 \times S_3$ (complement $\langle r^2, s \rangle \cong S_3$), extension (II) splits, and $[\mathfrak{g}] = 0$ on every base. The honest dividing line is non-splitting of (II), which on the n-lot ring is precisely $4 \mid n$ (yes for $n = 4, 8, 12, 16, \ldots$; no for $n = 6, 10, 14, \ldots$ and all odd n).

The split test was run for $n = 3, \ldots, 16$ and the arithmetic core – the centre table, the explicit complement for $4 \nmid n$, the centre-mismatch obstruction for $4 \mid n$, and the $D_6 \cong \mathbb{Z}/2 \times S_3$ reconciliation – is machine-checked in Lean, axiom-free (no `propext`, `sorryAx`, or `native_decide`).

6 Coupling is rare

Corollary 10 makes coupling a finite search: enumerate the commuting pairs in $\mathrm{Out}K$ and test whether any has no commuting lift in $\mathrm{Aut}K$. We ran this over a battery of nilpotency-class-two groups built from explicit multiplication tables, computing Aut, Inn, Out by brute force and cross-checking $|\mathrm{Aut}K|$ against known values to confirm the engine returns the *complete* automorphism group (a missed automorphism could fake a "no commuting lift"). This is *computed*, and over the listed family only – not a census of all groups of a given order.

K	$\lvert\mathrm{Aut}K\rvert$	$\lvert\mathrm{Inn}K\rvert$	$\lvert\mathrm{Out}K\rvert$	couples via (I)?
$D_4,\ Q_8$	8, 24	4	2, 6	no (split)
$\mathrm{Heis}(\mathbb{F}_3)$	432	9	48	no (split)
$M_{16},\ C_4 \rtimes C_4$	16, 32	4	4, 8	no
$\mathrm{ES}^+_{32} = 2^{1+4}_+$	1152	16	72	no (split)
$\boxed{D_4 \times C_2}$	64	4	16	**yes (30 pairs)**
$Q_8 \times C_2$	192	4	48	yes (54 pairs)

Only $D_4 \times C_2$ and $Q_8 \times C_2$ couple; every group whose extension (I) splits decouples, as Corollary 10 demands. The instructive rows are M_{16} and $C_4 \rtimes C_4$: their extension (I) need not split, yet they carry *no* coupled band. Non-splitting is genuinely weaker than coupling – exactly the gap flagged in Corollary 10: one further needs a *commuting* outer pair whose lifts cannot be made to commute. Coupling therefore requires two independent conditions to hold at once: a non-split automorphism tower *and* two commuting outer involutions with no commuting lifts. The richest composition failures live in this thin slice.

7 Consequences

For a torus-shaped supply chain – two independent, commuting, reversible operations – the two stages of the nonabelian inventory obstruction cease to be independent. The band-to-strictification connecting map is given exactly by a commutator of automorphism lifts (Theorem 8), so a nonzero band *forces* a nonzero degree-two class. The same engine, applied to the central extension, shows the inventory class itself couples; and it locates both phenomena sharply (the n-lot ring fails exactly when $4 \mid n$, Theorem 12) and rarely (a handful of structure groups across an enumerated family, §6).

For the grant's Applications pillar the reading is concrete. Global stock-keeping on a cyclic, reconvergent logistics network can fail *compositionally*, with no local remedy, but only when the stock fibre carries genuine gauge structure – a relabelling group with non-split automorphism tower – and only on a base rich enough to hold a degree-two class. Plain fungibility (symmetric-group fibres) never triggers it; the economically meaningful regime is allocation defined up to a gauge group. That the dangerous structure groups form a thin, explicitly identifiable slice turns a potential liability into a design criterion: one can read off, from the automorphism tower of the stock type alone, whether a two-cycle in the network can obstruct reconciliation.

Open edges. Three are worth naming, each a clean next target rather than a gap in what is claimed here. *(1) Minimality.* The rarity enumeration covers a listed family, not a census; whether $D_4 \times C_2$ is the unique minimal coupler over *all* groups of order ≤ 16 is open. *(2) Higher genus.* The δ-commutator lemma is special to $\mathbb{Z}^2$ only through the two-generator one-relator Koszul resolution; a genus-g base would replace the single relator by a product of commutators, and the class by a sum

of commutators of lifts – a natural extension of §3. *(3) Nonabelian kernel.* The class-two hypothesis makes $\mathrm{Inn}K$ abelian, so δ lands in ordinary H^2; dropping it replaces the clean connecting map by an Eilenberg–Mac Lane obstruction valued in a torsor of centres, genuinely harder and not needed for the inventory lane.

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